Business Analysis and Reporting

Kayden Padayachee

Top-Performing Region

The East region is the highest-performing region in terms of revenue, generating approximately 270,000. It is followed closely by the North region with 260,500, while the South region contributes 225,000. The West region is the lowest-performing area with 163,000 in total revenue. This indicates that most of the company's sales activity is concentrated in the East and North regions, while the West region may require additional attention to improve performance.

2. Highest Revenue-Generating Products

The analysis shows that Phones are the highest revenue-generating product, contributing around 440,000 in total revenue. This is significantly higher than all other products. Laptops follow with 180,000, while Tablets generate 150,000. Furniture products such as Desks and Chairs contribute much lower revenue, at 75,000 and 73,500 respectively. This clearly shows that electronics, especially phones, are the main drivers of revenue in the business.

3. Sales Trends Over Time

Sales show a steady upward trend across the four months analysed. January recorded the lowest revenue at 167,500, followed by an increase in February to 200,500. March shows a significant rise to 271,500, and April reaches the highest revenue at 279,000. This consistent growth indicates improving sales performance over time, possibly due to increased demand, better marketing strategies, or improved operational efficiency.

4. Top-Performing Salespeople

David is the top-performing salesperson, contributing approximately 270,000 in total revenue. John follows closely with 260,500, while Mary generates around 216,000. Sarah records lower performance at 163,000, and there is also an "Unknown" category contributing around 21,000. Overall, David and John are the strongest contributors to sales performance, while others may require support or improvement strategies.

5. Business Recommendations

Based on the analysis, several recommendations can be made to improve business performance. Firstly, the company should focus on increasing sales of high-performing products such as phones and laptops, as they generate most of the revenue. Secondly, targeted marketing strategies should be implemented in the West region to improve its low performance compared to other regions. Thirdly, training and performance support should be provided to lower-performing salespeople such as Sarah and the “Unknown” category to improve overall efficiency. Lastly, the company should consider reducing focus on low-revenue products like chairs and desks or bundling them with higher-value items to improve sales.

Report of my Findings

Environment Setup

The project was developed using Ubuntu (via WSL), which allowed me to work in a Linux-based environment while still accessing my Windows files. To access my project files, I first had to navigate to the correct directory using the cd command. Since the project was stored on my Windows drive, I used the /mnt/c/Users/kpada/OneDrive - ADvTECH Ltd/Desktop/sc_KaydenPadayachee_2026 path to access it from Ubuntu. So this part is important to also execute the pipeline properly.

```
kayden@kayden:~$ cd "/mnt/c/Users/kpada/OneDrive - ADvTECH Ltd/Desktop/sc_KaydenPadayachee_2026"
kayden@kayden:/mnt/c/Users/kpada/OneDrive - ADvTECH Ltd/Desktop/sc_KaydenPadayachee_2026$ cd "First Project"
```

Creating Project Files

I needed to make a and organise folders, so in ubuntu I used the commands mkdir to create the folders that were mentioned in the projected descriptions, like data, scripts, reports and output. Then I proceeded to create the process_data.py and run_pipeline.sh files in the output folder by using the commands touch and the file name, this automatically made the files in the output folder, then I navigated to were the python script was and opened it with pycharm to edit and write the script. I did this because pycharm highlights mistakes and shows errors were I went wrong. Then I used the command nano to edit the pipeline to edit and write the script for thre run_pipeline.sh.

Virtual Environment

I created a virtual environment known as venv, and this helps with isolating my project dependencies. So, it helps with ensuring that the libraries I used in the project do not interfere with other projects on my computer or system.

I used the command source venv to install it and I only need to do it once, because it creates a folder in my project folder. So, this helps with keeping the correct versions of the libraries like the panda, so it helps with making project more reliable.

Panada

I installed Pandas within the virtual environment because it is good for data processing and analysis in python.

Pip install pandas, but I had to update sudo and then sudo apt install python3 then I could install panda.

Panda provides powerful tools for working with structured data, such as reading CSV files, cleaning data and doing the aggregation that was used in my project.

Sudo stands superuser do, it's a command used in Linux, that lets you run a command with administrator (root) privileges.

ETL Process

The ETL (Extract, Transform, Load) pipeline was implemented in the Python script. During the extraction phase, the raw sales dataset was loaded from a CSV file located in the data directory. The transformation phase involved several data cleaning steps to ensure data quality. Duplicate records were removed to prevent inaccurate calculations. Missing values in text fields were filled with a default value and standardized for consistency. The quantity column was converted to a numeric format, and missing values were handled using a more advanced approach. Specifically, missing quantities were replaced using the average quantity per product, and the result was rounded to the nearest whole number. This ensured realistic and meaningful values rather than simply defaulting to zero. Any remaining missing values were filled with zero as a fallback. Dates were also cleaned by converting them into a standard datetime format and removing invalid entries.

```
/home/kayden/.hushlogin file.
kayden@kayden:~$ cd "/mnt/c/Users/kpada/OneDrive - ADvTECH Ltd/Desktop"
kayden@kayden:/mnt/c/Users/kpada/OneDrive - ADvTECH Ltd/Desktop$ cd sc_KaydenPadayachee_2026
kayden@kayden:/mnt/c/Users/kpada/OneDrive - ADvTECH Ltd/Desktop/sc_KaydenPadayachee_2026$ cd "First Project"
kayden@kayden:/mnt/c/Users/kpada/OneDrive - ADvTECH Ltd/Desktop/sc_KaydenPadayachee_2026/First Project$ cd project
kayden@kayden:/mnt/c/Users/kpada/OneDrive - ADvTECH Ltd/Desktop/sc_KaydenPadayachee_2026/First Project/project$ ./script
s/run_pipeline.sh
Starting the pipeline
Project directory: /mnt/c/Users/kpada/OneDrive - ADvTECH Ltd/Desktop/sc_KaydenPadayachee_2026/First Project/project

Activating virtual environment...

Running process_data.py...
2026-04-17 09:47:03 | start | success | Loading data from /mnt/c/Users/kpada/OneDrive - ADvTECH Ltd/Desktop/sc_KaydenPad
ayachee_2026/First Project/project/data/Messy_Sales_Data.csv
2026-04-17 09:47:03 | load data | success | 52 rows loaded
2026-04-17 09:47:03 | removing duplicates | success | 2 removed
2026-04-17 09:47:03 | quantity cleaning | success | All missing quantities filled using product averages
2026-04-17 09:47:03 | cleaned data | success |
2026-04-17 09:47:03 | date cleaning | success | 0 invalid dates removed
2026-04-17 09:47:03 | features | success |
2026-04-17 09:47:03 | clean file saved | success | /mnt/c/Users/kpada/OneDrive - ADvTECH Ltd/Desktop/sc_KaydenPadayachee
_2026/First Project/project/output/clean_sales.csv
2026-04-17 09:47:03 | aggregation phase | success | All reports created
DONE - Logs saved to CSV
```

Transformation

Additional transformations were performed to create new fields that support analysis. A revenue column was calculated by multiplying quantity by price, and a month column was extracted from the date field. These transformations made the dataset more suitable for business analysis. The cleaned data was then saved as a new CSV file in the output directory.

Aggregation

The pipeline also included a data aggregation phase, where the dataset was grouped to generate summary reports. These included sales by region, sales by product, monthly revenue, and salesperson performance. Each of these summaries was exported as separate CSV files, making it easier to analyse specific aspects of the data.

Logging

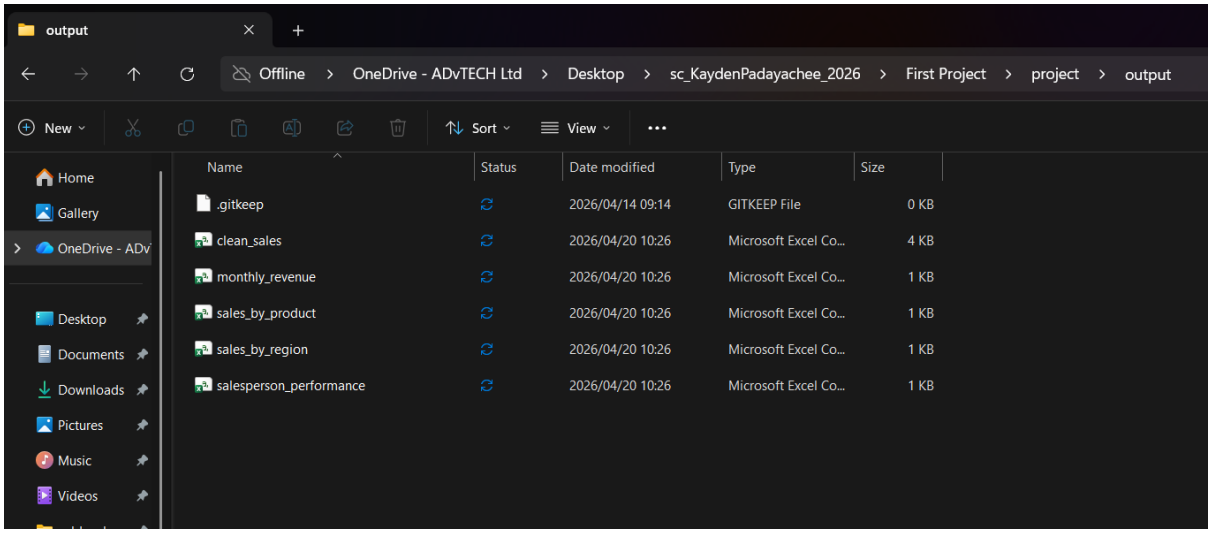
A logging system was implemented within the Python script to track the execution of each step. The logs included timestamps, step names, status indicators, and additional details. This feature improves debugging and ensures that any issues in the pipeline can be easily identified and resolved.

```
# logging function
def log(step, status, details=""):  Kayden
    entry = {
        "timestamp": datetime.now().strftime("%Y-%m-%d %H:%M:%S"),
        "step": step,
        "status": status,
        "details": details
    }
    logs.append(entry)
    print(f"{entry['timestamp']} | {step} | {status} | {details}")

# start
log( step: "start",  status: "success",  details: f"Loading data from {DATA_PATH}")

# load data
try:
    df = pd.read_csv(DATA_PATH)
    log( step: "load data",  status: "success",  details: f"{len(df)} rows loaded")
except Exception as e:
    log( step: "load data",  status: "error", str(e))
    raise
```

Automation

To automate the process, a Linux shell script named run_pipeline.sh was created. This script activates the virtual environment if it exists, runs the Python processing script, and checks whether all expected output files have been generated. It then displays a success or failure message based on the results. This automation ensures that the entire pipeline can be executed efficiently with a single command and reduces the risk of human error.

After processing the data, the cleaned dataset was imported into Microsoft Excel for analysis. Pivot tables were created to summarise key metrics, including sales by region, top-performing products, monthly revenue trends, and salesperson performance. Visualisations such as bar charts, line charts, and pie charts were created to present the data clearly. All visuals were combined into a single dashboard, allowing for an easy-to-understand overview of business performance.



Pivot Tables and Dashboard Analysis

After completing the data cleaning and transformation process, the final dataset was imported into Microsoft Excel for analysis. Pivot tables were used for summarising and exploring the data because they allow large datasets to be quickly reorganised into meaningful business insights without altering the original dataset.

Several pivot tables were created to address key business questions. A pivot table analysing sales by region provided a clear summary of total revenue generated across different geographic areas, making it easy to identify the best-performing and weakest-performing regions. Another pivot table focused on product performance, summarising total revenue and quantity sold per product category. This helped highlight which products contributed most significantly to overall sales and which products had lower demand.

A separate pivot table was used to analyse monthly revenue trends, which grouped the data by month and displayed total revenue over time. This made it possible to observe seasonal patterns and fluctuations in sales performance. In addition, a pivot table analysing salesperson performance was created to compare total revenue generated by each salesperson, allowing identification of top-performing individuals within the sales team.

To improve interpretation and communication of findings, these pivot tables were supported by visualisations such as bar charts, line graphs and pie charts. These charts made it easier to compare categories briefly and identify trends that might not be immediately obvious in raw form.

All pivot tables and visualisations were combined into a single Excel dashboard. The dashboard provides a consolidated view of key performance indicators, including regional performance, product profitability, monthly sales trends, and salesperson contributions. This allows stakeholders to quickly interpret business performance without needing to interact directly with the raw dataset.

Row Labels	Sum of revenue
East	270000
North	260500
South	225000
West	163000
Grand Total	**918500**

Row Labels	Sum of revenue
Chair	73500
Desk	75000
Laptop	180000
Phone	440000
Tablet	150000
Grand Total	**918500**

Row Labels	Sum of revenue
Jan	**167500**
Feb	**200500**
Mar	**271500**
Apr	**279000**
Grand Total	**918500**

Row Labels	Sum of revenue
Chair	73500
Desk	75000
Laptop	180000
Phone	440000
Tablet	150000
Grand Total	**918500**